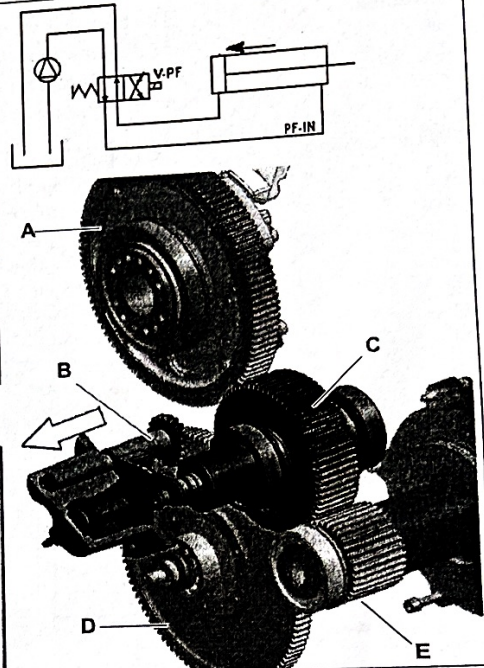
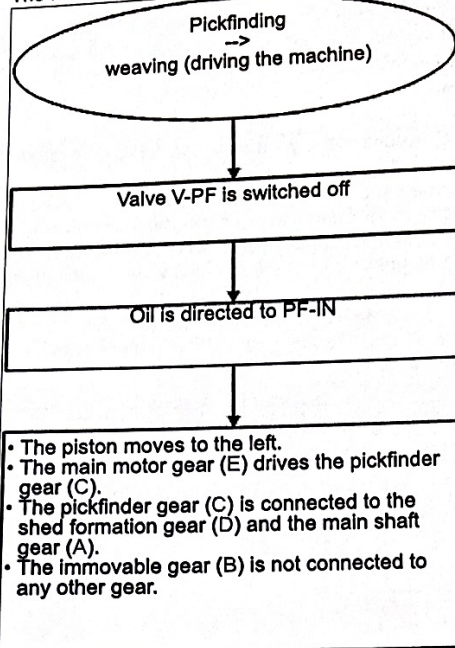


Weaving (driving the machine)

Process

The following illustration shows the weaving process:



PXWP and PXPFP

General

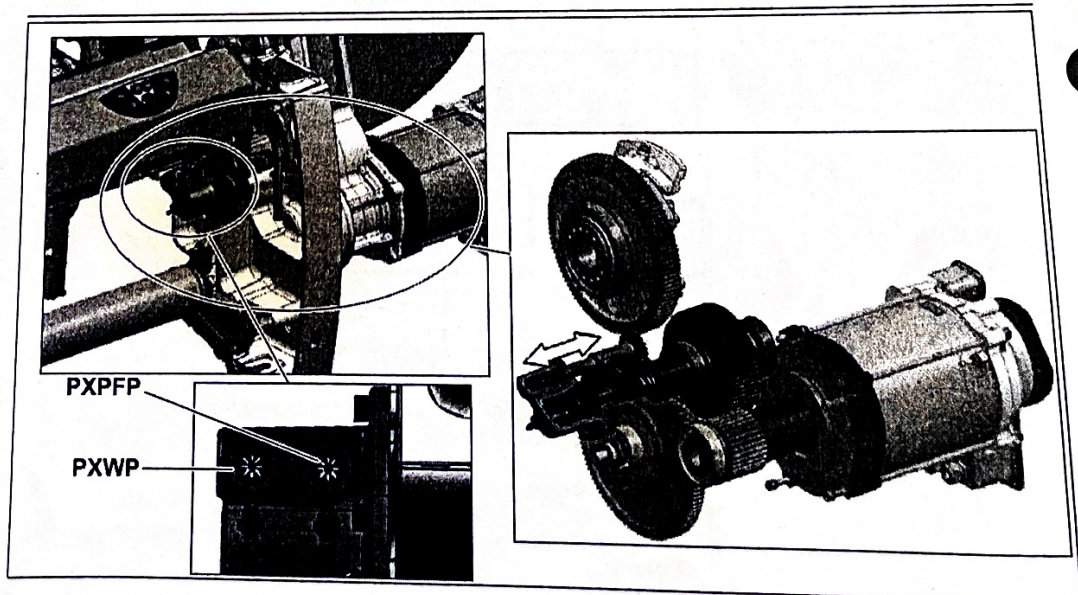
Two proximity switches are used to determine the position of the pickfinder (weaving, pickfinding):

- PXWP (PX weaving position);
- PXPFP (PX pickfinding position).

See also:

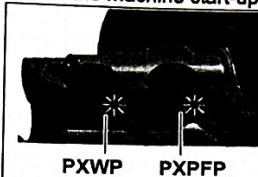
- Pickfinding [19]
- Weaving (driving the machine) [20]

Location



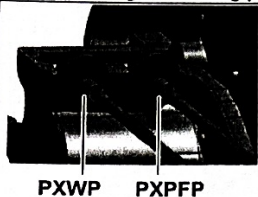
PXWP

This proximity switch checks the weaving position.
Before the machine start-up, during weaving and during slow motion, this PX should be activated.



PXPFP

This proximity switch checks the pickfinding position:
Before starting and during pickfinding, this PX may not be activated.



Let-off

General

The let-off comes in 2 versions:

- without speed change gear box;
- with speed change gear box.

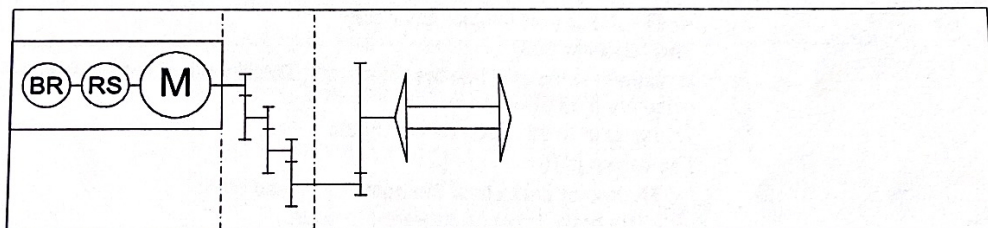


A speed change gear box offers more pick density and let-off speed options to ensure an optimal working of the let-off.

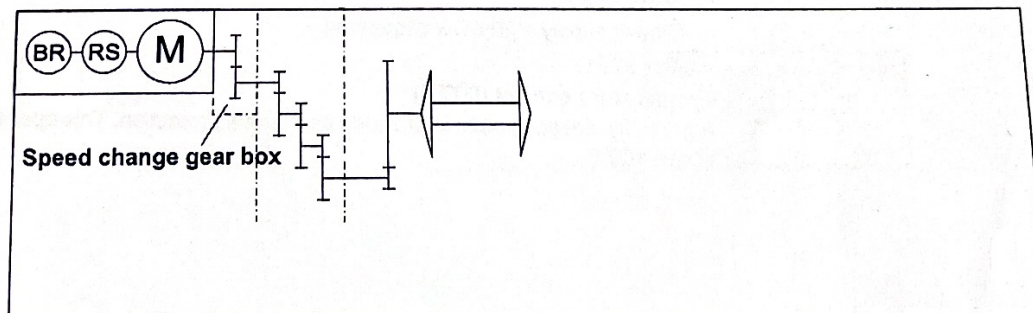


The electrical connection is the same for both versions. The version is determined in the MID settings.

Without speed change
gear box



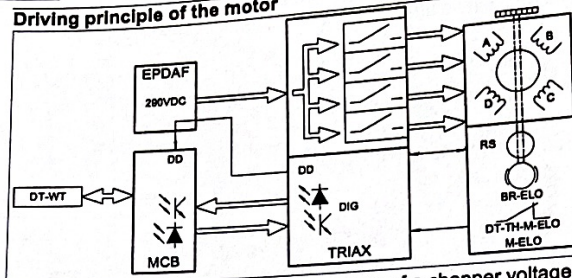
With speed change
gear box



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The let-off motor(s) M-
ELO-*
(* = 1, 2 and FA)

Driving principle of the motor



- The stator coils are controlled by means of a chopper voltage from the TRIAX-print control modules.
- The power supply to the TRIAX print for the power stages is 290V DC.
- The current in the different motor phases is monitored by means of current sensors. If 1 measurement is incorrect, the control modules for the motor concerned are switched off by means of the Drive Disable (DD) signal.



Depending on the machine configuration, three different let-off motors can be used, i.e.:
These are:

- ELO1
- ELO2
- ELO-FA



Depending on the machine configuration, 2 TRIAX prints can be present, i.e.:

- the TRIAX print (in the main control box)
- the T_TRIAX print (in the TRIAX control box)

The motor (A, B, C and D)

The motor contains 4 double stator coils.

=> $R = 2.3 \Omega$ (per double stator coil)

The resolver (RS)

A resolver is mounted on the rotor shaft. The signals of the resolver are used to continuously calculate the motor speed.

This data is used to control the motor.

The brake (BR)

- The motor brake is of the normally closed type:
 - The brake must be powered to open.
 - The brake keeps the warp beam in place while power is OFF or in extended downtime situations.
- Power supply = 26V DC choppered
- $R = \pm 26 \Omega$

Temperature control (DTTH)

A normally closed bimetal is provided as thermal protection. This opens when the temperature rises above 100°C.

The warp tension sensors

There are 2 types of warp tension sensors:

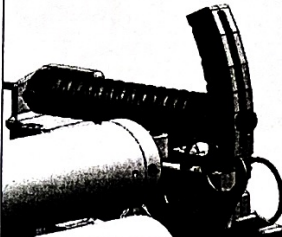
- TSF;
- TSW.

The table below shows when the 2 types are used.

Machine type	TSF	TSW
single beam	standard 1 sensor	optional 1 sensor
double beam	not possible	2 sensors
fancy on machine	2 sensors	not possible
fancy behind machine	3 sensors	

If multiple sensors are present on the machine, they are all the same type.

TSF sensor



TSW sensor



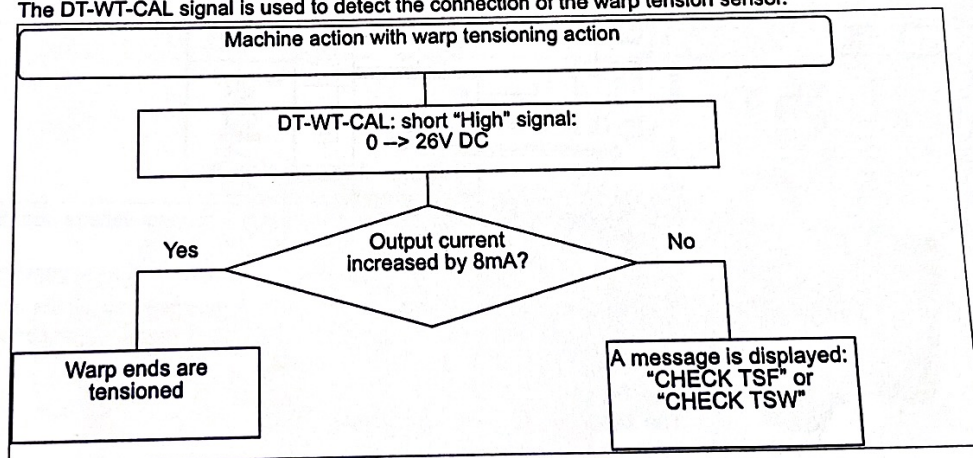
- The warp tension sensor converts the warp tension to an analogue DC current (4-20 mA).
- The warp tension sensors must be calibrated before accurate measurements are possible.
- Power supply for the warp tension sensors = +26V DCS

Warp tension limit values

- Warp tension and warp tension limit values are set using the machine terminal.
- If the warp tension limit values are exceeded, the machine stops.

Monitoring the sensors

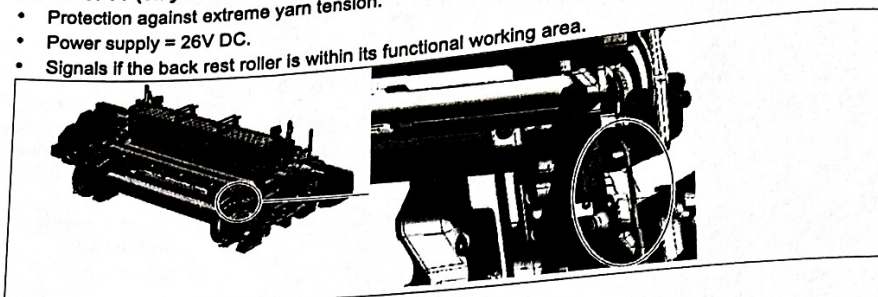
The DT-WT-CAL signal is used to detect the connection of the warp tension sensor.



OptiMax-i

PX-LO-SFTY (only on BLI or BLIDI back rest)

- Protection against extreme yarn tension.
- Power supply = 26V DC.
- Signals if the back rest roller is within its functional working area.

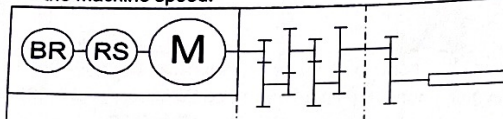


Take-up motion

General

The TRIAX print microprocessor controls the ETU motor on the basis of:

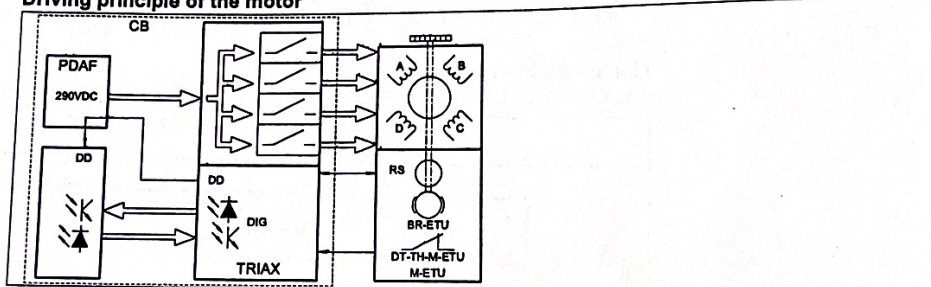
- the set pick density;
- the machine speed.



The ETU motor drives the sand roller through a gear train.

The M-ETU take-up motor

Driving principle of the motor



- The stator coils are controlled by means of a chopper voltage from the TRIAX print control modules.
- The power supply to the TRIAX print for the power stages is 290V DC.
- The current in the different motor phases is monitored by means of current sensors. If 1 measurement is incorrect, the control modules for the motor concerned are switched off by means of the Drive Disable (DD) signal.

The motor (A, B, C and D)

The motor contains 4 double stator coils.

=> $R = 2.3 \Omega$ (per double stator coil)

The resolver (RS)

A resolver is mounted on the rotor shaft. The signals of the resolver are used to determine the angle position of the motor.

This data is used to control the motor.

The brake (BR)

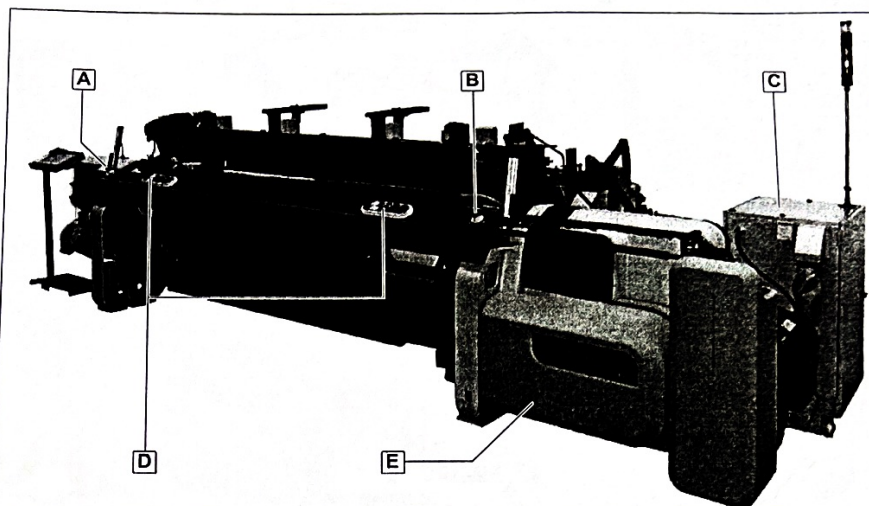
- The motor brake is of the normally closed type.
 - The brake must be powered to open.
 - The brake keeps the cloth roll in place while power is OFF or in extended downtime situations.
- Power supply = 26V DC choppered.
- $R = \pm 26 \Omega$

Temperature control (DTTH)

A normally closed bimetal is provided as thermal protection. This opens when the temperature rises above 100°C.

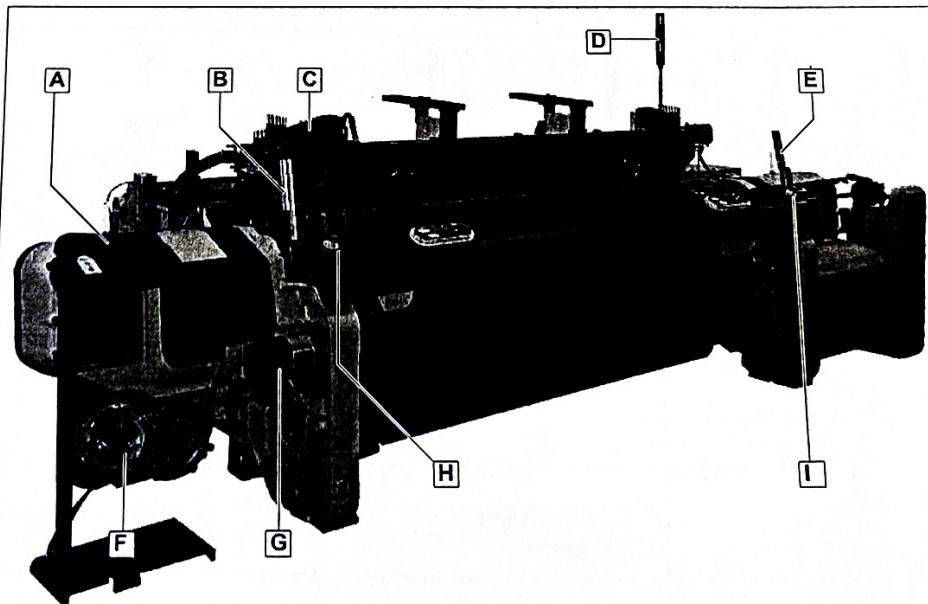
Overview of the weaving machine

Overview of weaving machine 1



A. ES-LHS: emergency stop	B. ES-RHS: emergency stop
C. CB: main Control Box	D. PB-*: push buttons (* = STRT, STP, PFF, PFR, SMF and SMR)
E. MM: Main Motor (Behind cover)	

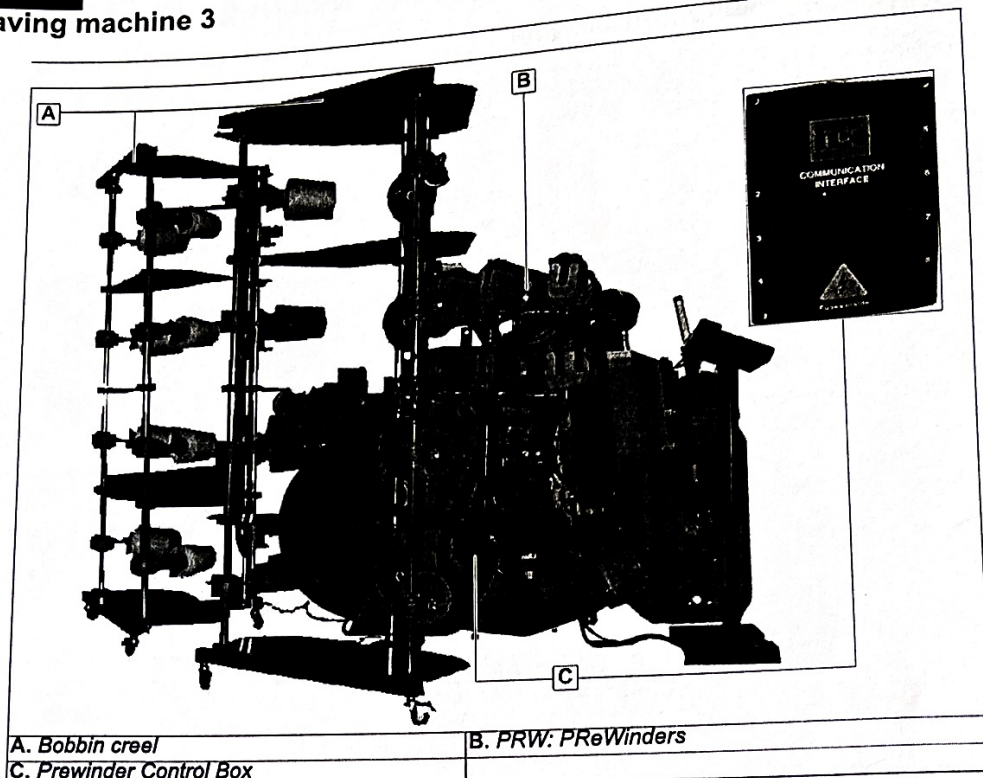
Overview of weaving machine 2



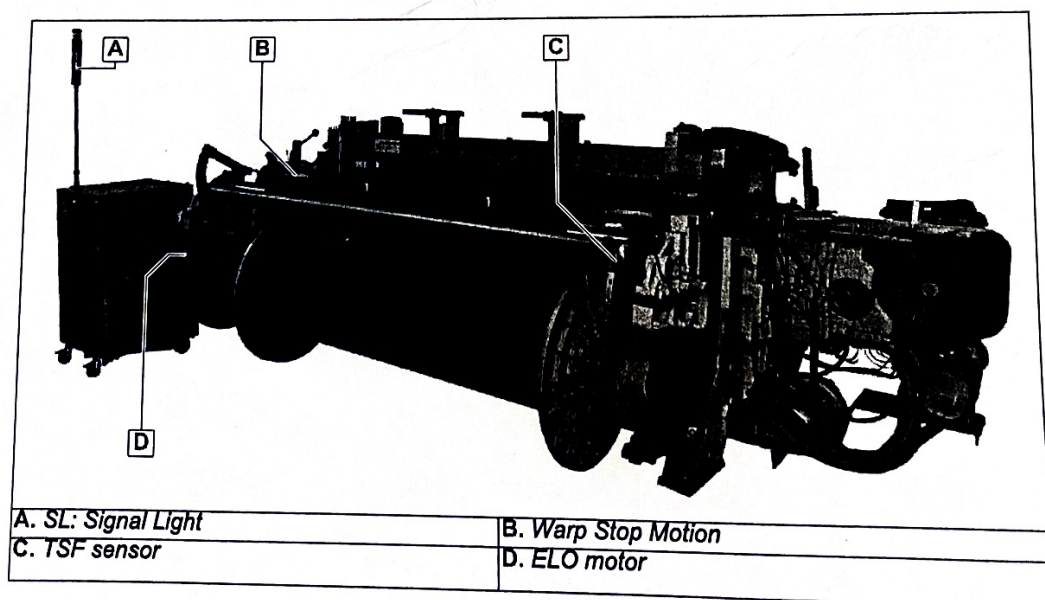
A. Machine terminal	B. Safety light curtain (Transmitter)
C. Quickstep	D. Signal light
E. Safety light curtain (Receiver)	F. Aspirator
G. ETU motor	H. WL_LHS: Warning Light
I. WL_RHS: Warning Light	

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Overview of weaving machine 3

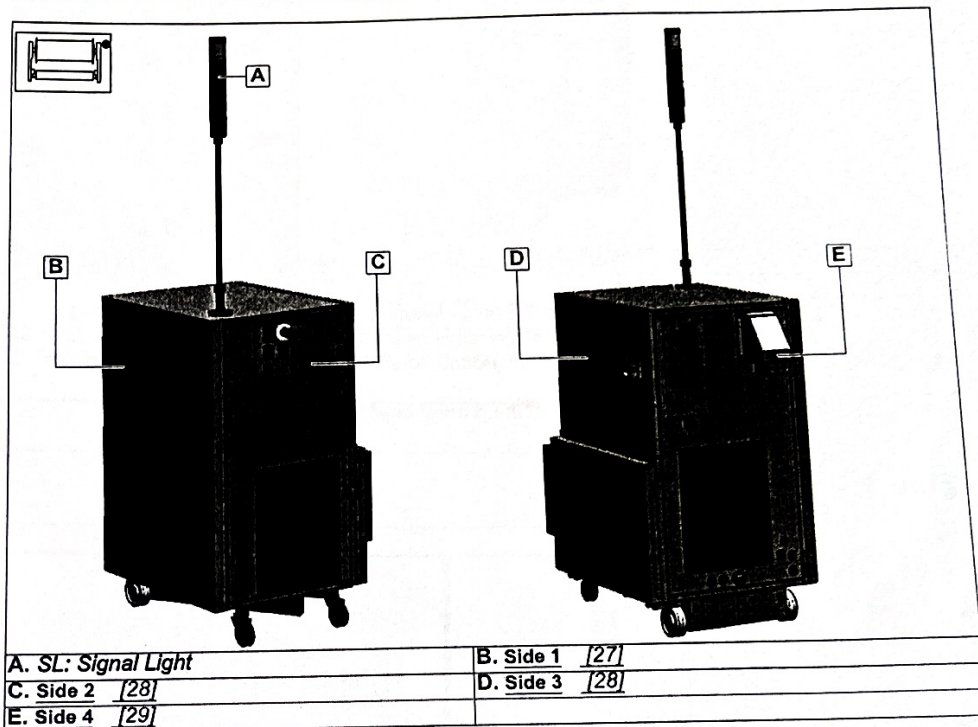


Overview of weaving machine 4

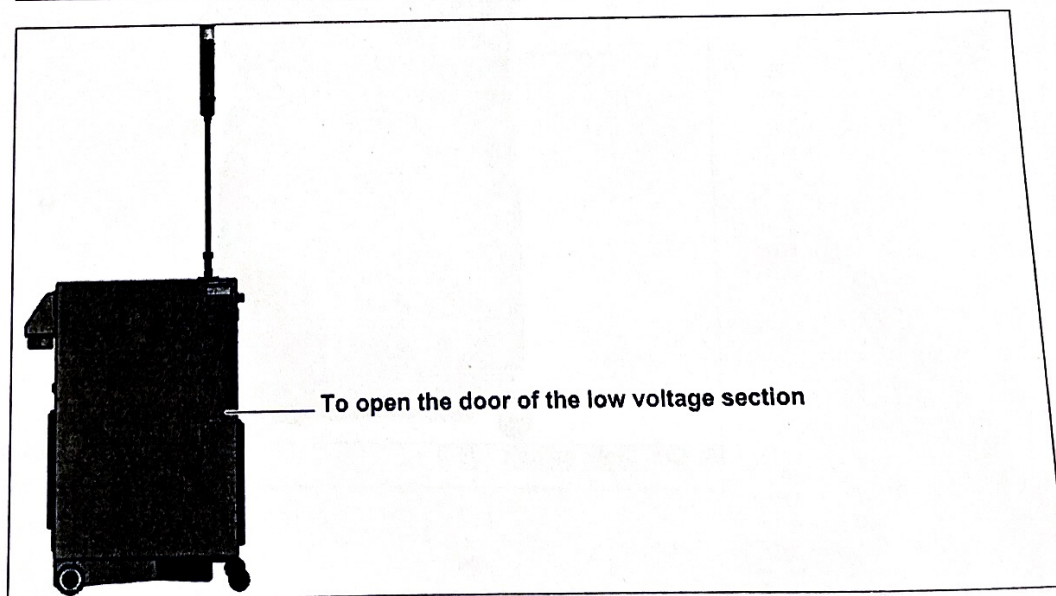


Main control box (CB) > Parts on the outside of the main control box

Overview of the outside of the main control box

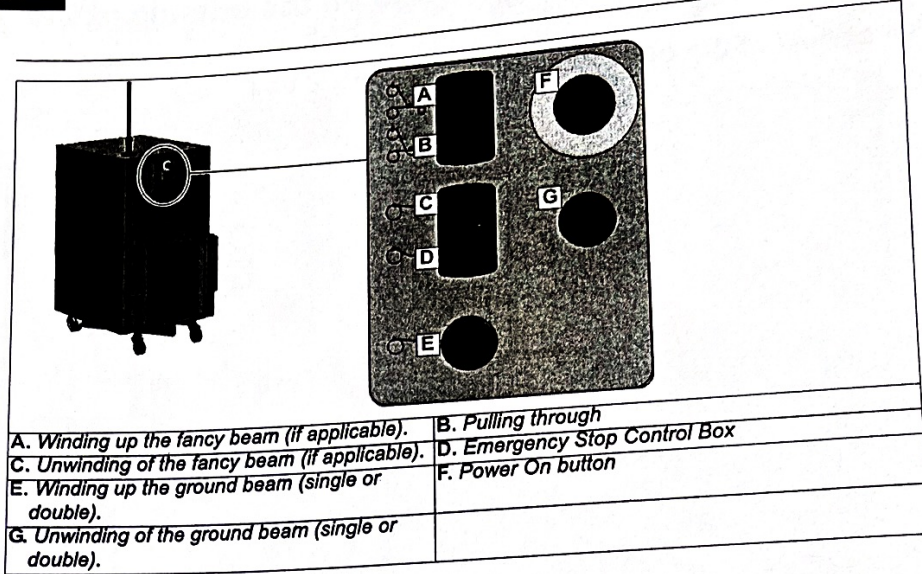


Side 1

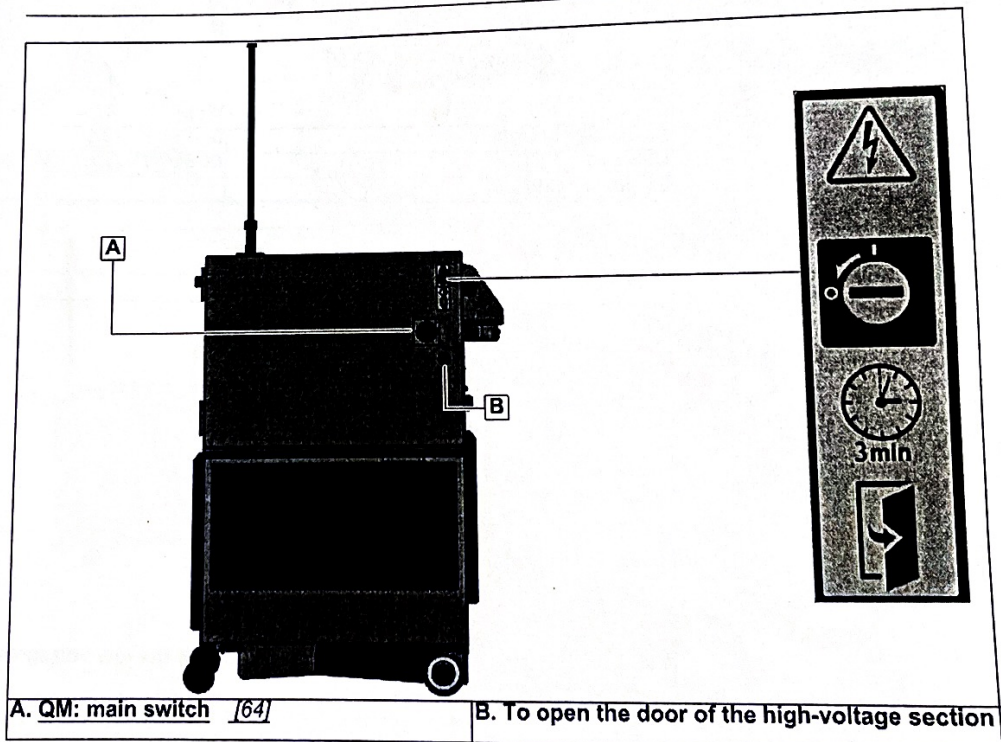


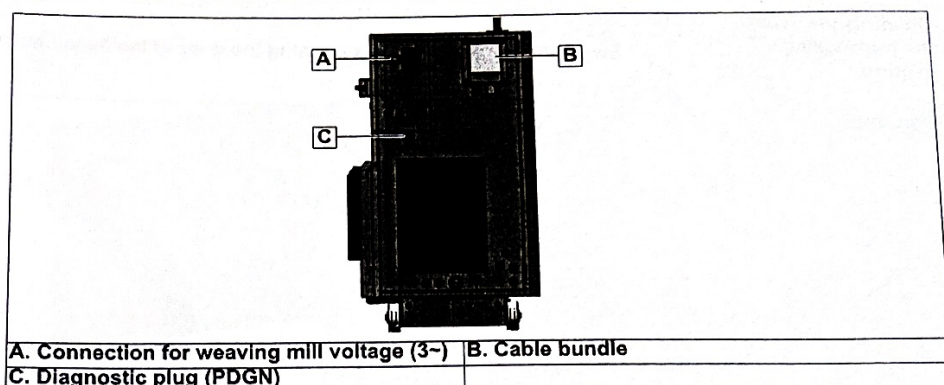
Side 2

OptiMax-i



Side 3





Main control box (CB) > High-voltage section

Opening the door to
the high-voltage
section

See Side 3 [28].



Switch off the mains before opening the door of the main control box.

Contents

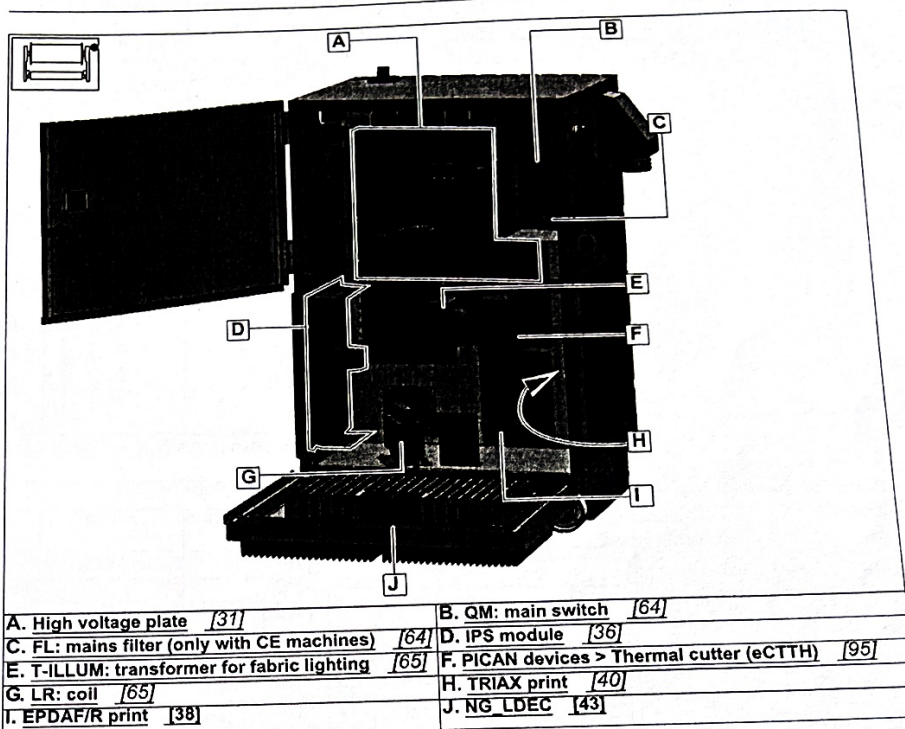
- General [30]
- Components [31]
- High voltage plate [31]
- High voltage plate > Ratings for fuses and safety switches [34]
- High voltage plate > INRUSH print [36]
- IPS module [36]
- EPDAF/R print [38]
- TRIAx print [40]
- NG LDEC [43]
- NG LDEC [43]
- NG LDEC > OBMM110 print [52]
- NG LDEC > RBMM110 print [56]
- NG LDEC > CBMM110 print [60]
- NG LDEC > EBMM1020 print [62]
- QM: main switch [64]
- FL: mains filter (only with CE machines) [64]
- LR: coil [65]
- T-ILLUM: transformer for fabric lighting [65]

General

Electrostatic
discharges

The printed circuit boards contain integrated circuits that are sensitive to electrostatic discharges:

- Put the printed circuit boards away in special boxes.
- Do not place the printed circuit boards on a metal surface, to avoid short-circuiting the batteries on the board.
- Do not touch the printed circuit boards unnecessarily and handle them with care to avoid permanent damage to the circuits.



High voltage plate

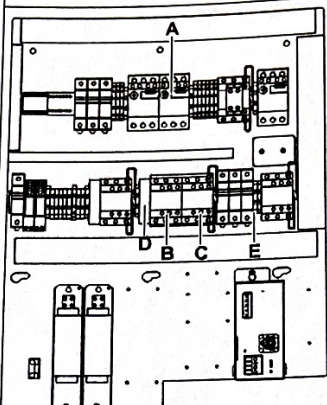
Standard parts

Illustration	Component	Name	Description
	A	F-OUT-1/2/3 (1)	safety fuse
	B	INRUSH	High voltage plate > INRUSH print [36]
	C	K-M-ASP	motor switch
	D	K-MM	main motor
	E	K-SFTY	machine protection
	F	L11/L12/L13	terminals
	G	L21/L22/L23	
	H	Q-M-ASP (1)	safety switch
	I	Q-M-LU (1)	oil pump motor
	J	X1	earth terminal
	K	X3	main motor
	L	X4	aspirator motor
	M	X6	
	N	X10	
	O	XK (1, 4, 5 and 6)	connection terminals. See Batching motion.
	P	Z-M-ASP	filter

(1): see High voltage plate > Ratings for fuses and safety switches [34].

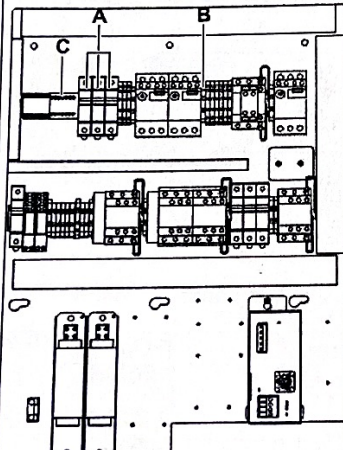
OptiMax-i

Outside cam motion
leveling components

Illustration	Component	Name	Description
	A	Q-M-LVL (1)	safety switch
	B	K-M-LVL-ON	motor switch (status ON)
	C	K-M-LVL-OFF	motor switch (status OFF)
	D	Z-M-LVL	filter
	E	X2	earth terminal

(1): see High voltage plate > Ratings for fuses and safety switches [34].

Parts for the batching
motion

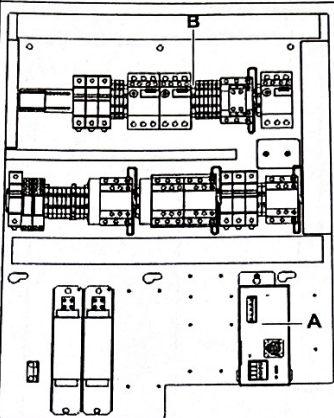
Illustration	Component	Name	Description
	A	F-OUT-1/2/3 (1)	fuses
	B	X4	earth terminal
	C	XK (1, 4, 5 and 6)	connection terminals. See Batching motion.

(1): see High voltage plate > Ratings for fuses and safety switches [34].

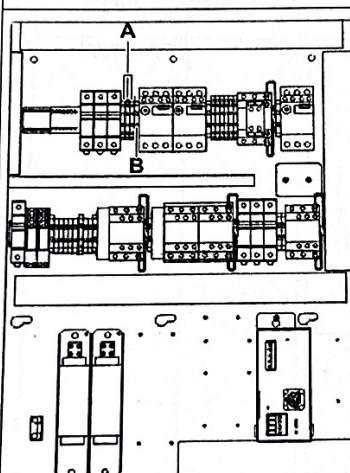
**LET'S GROW
TOGETHER**

Thermal cutter
components

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Illustration	Component	Name	Description
	A	eCTTH	thermal cutter
	B	X4	earth terminal

Components for the
jacquette

Illustration	Component	Name	Description
	A	L21/L22/L23	terminals
	B	X6	earth terminal

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Components for fabric
lighting

Illustration	Component	Name	Description
	A	F-T-ILLUM (1)	safety fuse
	B	COIL-IL-UP-1/2	choke upper lighting
	C	K-IL-LOW	motor switch
	D	K-IL-UP	lower lightening
	E	X9	upper lightening
	F	T-ILLUM	connection terminals + earth terminal transformer for fabric lighting

(1): see High voltage plate > Ratings for fuses and safety switches [34].

Components for fancy

Illustration	Component	Name	Description
	A	FFA-1/2/3(1)	fuses

(1): see High voltage plate > Ratings for fuses and safety switches [34].

High voltage plate > Ratings for fuses and safety switches

Fuses

Fuse	Value	Order Number
F-OUT-1/2/3	6 AM	N1020046
F-FFA-1/2/3	6 AM	N1020046
F-T-ILLUM	1.5ATQ	N1020120

Safety switches power
supply settings

Safety switch	Mains voltage	Current at 50 Hz	Current at 60 Hz
Q-M-ASP	380 V	1,4 A	1,5 A
	400 V		
	415 V		
	440 V		1,3 A
	460 V		

**LET'S GROW
TOGETHER** 

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Safety switch	Mains voltage	Current at 50 Hz	Current at 60 Hz
Q-M-LU	380 V	1,0 A	1,2 A
	400 V		1,1 A
	415 V		1,0 A
	440 V		
	460 V		
Q-M-LVL	380 V	4 A	4 A
	400 V	4,5 A	4,5 A
	415 V	2 A	
	440 V	2,2 A	
	460 V	2,4 A	2,5 A

OptiMax-i

Order numbers for
safety switches

Current safety switch	Order Number
0,25 – 0,4 A	N1022845
0,63 – 1 A	N1022846
1 – 1,6 A	N1022847
1,6 – 2,5 A	N1022848
2,5 – 4 A	N1022849
4 – 6,3 A	N1022850

High voltage plate > INRUSH print

Version

INRUSH-3: BE232433

General

This print contains three power resistors.

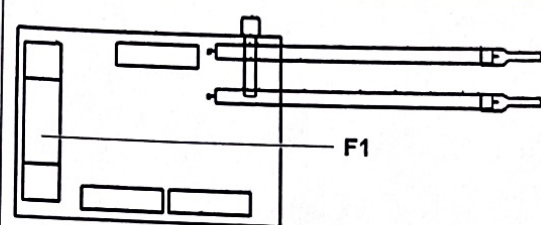
Function

This print is for the loading of the ELCOs of the NG_LDEC.
This limits the inrush current at machine start-up.

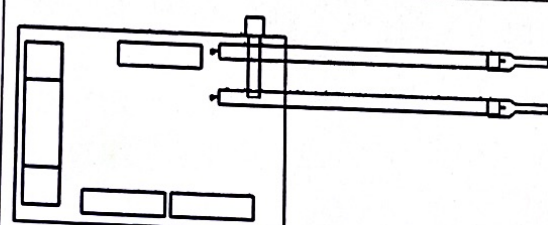
Signals

There are no LEDs on the print.

Fusing

Illustration	Fuse	Order Number	Value
	F1	N1020122	6,3 AT

Connections

Illustration	Connections
	KMM-1 KMM-2

IPS module

Name

Industrial Power Supply

Version

IPS-3: BE310253

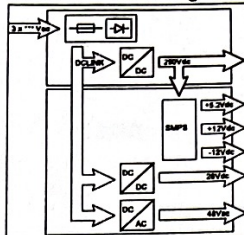
General

- This module is made up of two prints that form one unit.
- This is an air-cooled print. The cooling system of this print can be severely disrupted by dirt such as dust (dressing). As a consequence, to ensure the proper operation of the IPS module it is very important that the air circulation on and around the cooling plate of the print is kept optimal. The heatsink of the print should be blown clean regularly (with every gaiting).

Function

This module:

- distributes the voltage.



- checks the input voltage (AC-OK).

If the 3 phases are...	then...
OK (+12VDC)	an AC-OK signal is sent to the MCB print via the EPDAF print.
not OK (0V)	the microprocessor on the MCB print will not start. The error message "No communication with the machine" will appear on the machine terminal.

- checks all DC voltages (DC-OK).

If all DC voltages are ...	then...
OK (+12VDC)	a DC-OK signal is sent to the MCB print via the EPDAF print.
Not OK (0V)	one of the DC voltages is not present. If the 5V DC is not present either, the MCB print will not start.

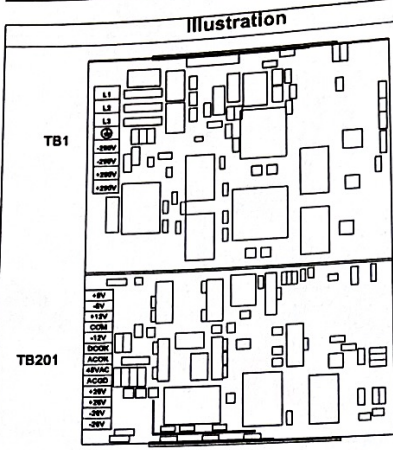
Signals

There are no LEDs on the print.

Fusing

Illustration	Fuse	Circuit	Order Number	Value
	F1	mains voltage	N1020116	8 AT
	F2	380 – 460V AC		
	F3			
	F201	48V AC	N1020117	3,15 AT

Connections

Illustration		Connector	Connections
		TB1	- mains voltage 290 VDC -290 VDC
		TB201	48 VAC 26 VDC -26 VDC 12 VDC -12 VDC 5 VDC -5 VDC - DCOK - ACOK

EPDAF/R print

Name

Enhanced Power Distribution And Fusing / Rapier

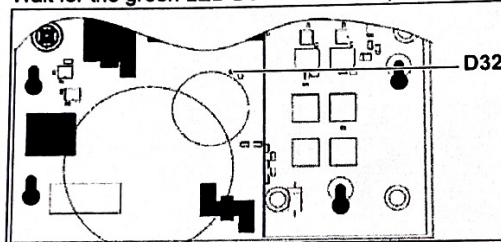
Version

EPDAF/R-1: BE234927

Safety

Before working on the print:

1. Set main switch QM to position -0-.
2. Wait for the green LED D32 to fade completely, because the capacitors discharge only slowly.

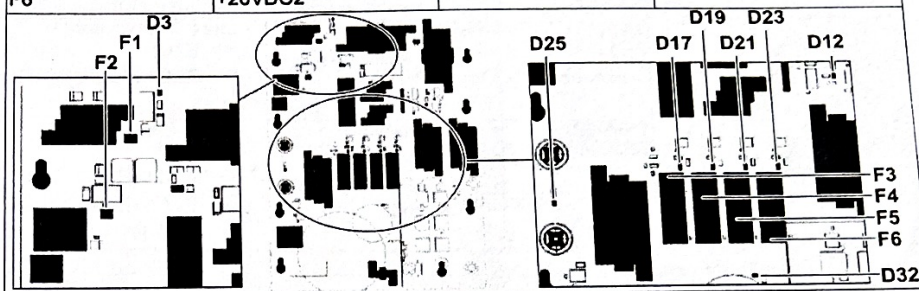


Function

This print:

- protects and distributes the voltages coming from the IPS module;
- connects monitoring and control signals through;
- connects the emergency stops;
- connects the warning lights;
- controls the power to the electric safety brake;
- connects and secures the power supply to the PICAN bus;
- connects the PICAN signals.

LED	Color	The LED is lit if...	
D3	green	48V AC	is present.
D12		PBR signal	
D25		+26V DC	
D32	yellow	+290V	is not present.
D17		+26VDCPICAN	
D19		+26VDCS	
D21	red	+26VDC1	
D23		+26VDC2	
Fuse	Circuit	Value	Order Number
F1	48V AC	300 mA at	
F2	41V DC		
F3	+26VDCPICAN	8 AT	N1020091
F4	+26VDCS		
F5	+26VDC1		
F6	+26VDC2		



Connections

Illustration	Connector	Connections
	EPDAF-01	- Power supply coming from the IPS module. - Emergency stop for jacquard.
	EPDAF-02	PICAN connections with MCB and devices.
	EPDAF-04	Power supply for IBR, MCB, SBPO and CBES and POWER ON button.
	EPDAF-05	Power supply (5V DC) coming from the IPS module.
	EPDAF-06	RHS emergency stop and warning light.
	EPDAF-07	Power supply for the main control box fan.
	EPDAF-08	Control signals to MCB.
	EPDAF-09	Not used.
	EPDAF-11	LHS emergency stop and warning light.
	EPDAF-12	- Power supply to the TRIAX print. - Power supply to prewinders. - Signals to the safety brake. - DD2 of the TRIAX print.
	EPDAF-13	- Power supply to the T_TRIAX print. - DD1 of the T_TRIAX print.
	EPDAF-14	Power supply for the IBR, the UEB_X, the UEB_Y, the ATB_X, the ATB_Y print, and the ATB_Z print.
	EPDAF-15	Power supply (26VDC; 290VDC) of the IPS print.
	EPDAF-16	